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TO THE OUTSIDER, it may seem that progress in bone marrow transplantation is slow in present years, because the days of impressive breakthroughs are over and clinical results tend to accumulate slowly. For the insider, however, the field is steadily advancing. Clinical bone marrow transplants are increasing in number. According to figures from the International Bone Marrow Transplant Registry of ACS/NIH, a total of 350 patients has been reported so far and the accession rate at present is 12 new cases per month. Transplant patients not reported to the registry are estimated at another 250, many of whom have been described in the literature. At least 40 transplant teams are known to be active in the field.

The three main indications for bone marrow transplantation are: severe combined immune deficiency (SCID), bone marrow aplasia, and leukemia. These three diseases themselves have little in common, but they share many of the difficulties encountered in the use of bone marrow grafts as a treatment, namely the lack of suitable compatible donors for many patients, the development of graft-versus-host disease (GVHD) in a majority of patients in whom a bone marrow graft has been established, and finally the prolonged period of immune deficiency that follows allogeneic bone marrow transplantation resulting in a high risk of fatal infectious complications.

BONE MARROW TRANSPLANTATION IN THE TREATMENT OF SEVERE COMBINED IMMUNE DEFICIENCY (SCID)

In the treatment of SCID the sole objective is to reconstitute the immune system and, if this succeeds, the patient seems to be permanently cured. From the point of view of bone marrow transplantation the disease is unique in that there are virtually no immunologic barriers and, therefore, conditioning of the patient is usually not required. SCID is a very rare disease and the experience of each transplantation team with this disease is limited. Therefore, it is gratifying that the International Bone Marrow Registry is now in the position to generate interesting trends in the results of transplantation of these patients, thus helping the transplanters to identify the problems requiring further study. Sixty-eight SCID patients who have undergone transplantation since 1968 have been reported by 18 transplant teams. The following conclusions are based on Bortin's report at this conference,[1] and on data provided to me by the registry.*

The subgroups of patients to be discussed are: (1) those treated with matched (mostly HLA-identical sibling) donor marrow, (2) those grafted with incompatible (mis-

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*Unedited and not approved by the registry's Advisory Committee on Bone Marrow Transplantation.

Table 1. Bone Marrow Transplantation in Severe Combined Immune Deficiency (Bone Marrow Registry Data)

HLA-Matched Family Donors			HLA-Mismatched Donors	
Take	16/16		21/23	
GVHD	11/16	3 severe (1 sibling)	18/21	7 severe
		4 moderate		5 moderate
		4 mild		6 mild
Survival > 40 months: 10/16			Anti-HLA or enhancing serum: 4/5 severe GVHD	
			Survival > 40 months: 4/23	

matched) marrow, and (3) patients given fetal liver cells. Although the reporting of takes and GVHD is done according to rigid criteria set by the Advisory Committee of the Bone Marrow Registry,[2] one may anticipate some differences in judgment and interpretation between the various reporting teams. Nonetheless, certain trends appear, which cannot be due to such variations (Table 1).

One conclusion is that an engraftment of the donor marrow is always seen in the matched group (16 cases) and nearly always in the mismatched group (21 out of 23 cases) regardless of the number of cells grafted. The latter varied between 10^6/kg and 1300×10^6/kg.

The second—more surprising—finding is that the incidence of GVHD in the matched group is 69% (64% when two of the cases with severe GVHD are disregarded because they were treated with matched marrow from nonsibling donors). This may seem to be in contrast with the generally held opinion that SCID patients are far less susceptible to GVHD following bone marrow transplantation than anemia and leukemia patients. However, the degree of GVHD is less in these patients: only one of nine who received matched sibling marrow and developed GVHD was graded as severe. On the other hand, five of these nine patients with GVHD did not survive and, although the causal relation between this mortality and the GVHD is not known, this mortality again falls in the range observed in transplanted aplastics and leukemics.[3,4]

With regard to the relationship between incidence and severity of GVHD and the number of bone marrow cells grafted, the registry data do not provide definite answers yet. In the matched transplant group there is an indication of a preventive effect of small graft sizes as compared to large ones: the incidence of GVHD in patients receiving less than 3×10^7 cells/kg is 2/5 compared with 7/8 for those receiving more than 10^8 cells/kg. GVHD occurred in 18 of 21 patients following engraftment of mismatched bone marrow. In many of these cases some kind of manipulation was applied to the bone marrow cells before grafting, or the recipients were given additional treatment aimed at mitigation of GVHD. Also a relatively large proportion of patients received the mismatched bone marrow via the i.p. route. Because of such heterogenicity this group allows no conclusions with regard to the influence of cell numbers. It is interesting that of the three cases *not* developing GVHD, two received small numbers of gradient prepared cells (survival, 14 and 76 months +) and the third case was given 9×10^6 unmodified bone marrow cells/kg i.v. (survival, 52 months +).

The important lessons from all these data are that (1) GVHD may occur more frequently in SCID patients receiving matched sibling marrow than was generally assumed; (2) there is no sound reason to graft excessive numbers of bone marrow cells into any SCID recipient, since this may increase the risk of GVHD developing; and (3) that in mismatched transplants the attempts at immunologic reconstitution with gradient-purified enriched stem cell concentrates are worthwhile to continue. These conclusions, tentative as they may be, are in principle supported by experimental evidence previously obtained in animal models.[5]

With regard to the effect of "enhancing sera" on the incidence and severity of GVHD, it is noteworthy that four of the five patients (of mismatched series) who were treated with anti-HLA or "enhancing sera" (Table 1) were reported to develop severe GVHD. Marquet and I now have strong evidence in rat models that the same sera that enhance the survival of organ allografts quite effectively are without effect and under certain conditions even aggravate the GVHD (Table 2).

Considerably less information has been accumulated on the grafting of fetal liver cells in the treatment of SCID (only nine cases reported). In sharp contrast to the findings with unmatched bone marrow, there is a high proportion of take failures with liver cells of fetuses of 12 weeks and older. An explanation is so far not available for this discrepancy, apart from the fact that Löwenberg[6] observed fetal liver hemopoietic stem cells to encounter a comparatively greater barrier in repopulating lethally irradiated allogeneic mouse recipients than was the case with allogeneic bone marrow. If any suggestion can be made from the human fetal liver data, it might be that all three patients reported to survive longer than 1 year were treated with liver cells from very young fetuses (5, 8, and 9 weeks). Although the cell numbers grafted were relatively small (2, 14, and 65 × 10^6/kg i.p.) a take was achieved in all three cases. Data recently collected by Buckley[7] also show a preponderance of young fetal donors among the successfully transplanted cases.

Table 2. Effect of Enhancing Serum in a Rat Model (5×10^7 BN Spleen Cells → (WAG/Rij × BN)F1 – 450 rads)

Treatment	Mean Survival
Protocol 1	
0.5 ml days 0, 2, 4, 6, 8	
None	23.4
NRS	25.9
ES*	23.0
Protocol 2	
0.5 ml days 7, 9, 11, 13, 15	
None	26.4
NRS	25.0
ES	19.4

Obtained from sensitized BN rats following accelerated rejection of a WAG cardiac graft. These sera induce effective enhancement of WAG allografts in BN recipients.

NRS, normal rat sera; ES, enhancing sera.

These registry data underline the disadvantage of the use of many different treatment procedures; few reliable conclusions can be drawn from the results accumulated during a period of 7 years. It is therefore encouraging that the majority of transplanters involved in the treatment of SCID have formed a club[8] under the auspices of the International Cooperative Group on Bone Marrow Transplantation in Man and of the International Society for Experimental Hematology and that the members of this club are making serious attempts to devise and adopt common clinical protocols.

BONE MARROW APLASIA AND LEUKEMIA

The main problems that were identified several years ago in the treatment of bone marrow aplasia and leukemia remain to be resolved. One that seems to be specific for leukemia is the significant incidence of relapse following bone marrow transplantation. Several attempts are being made to reduce this incidence by intensifying the leukemic eradication regimens prior to bone marrow grafting. Since this type of effort does not strictly belong to the transplantation field, it shall not be discussed here.

One problem that seems to be specific for aplasia is the failure of the graft to take in up to one-third of the cases. So far most of these patients are being conditioned with high-dose cyclophosphamide only, in contrast to leukemic patients in whom a combination of cyclophosphamide and whole-body irradiation has been employed during recent years. It appears that take failure can be partly prevented by the use of sufficiently large numbers of bone marrow cells (4×10^8/kg). This notion has been stressed for years by our group on the basis of results in several animal models and has now been substantiated by clinical data from Seattle.[9]

Two other problems are common to bone marrow transplantation in aplasia and leukemia.

(1) The absence of suitable donors for 50% or more of the candidates for bone marrow transplantation is a major problem. These patients are presently not being treated with bone marow grafts because the grafting of incompatible bone marrow results in fatal GVHD. In a few centers induction of temporary or partial engraftment is being attempted in selected aplastic patients by conditioning with ALG.[10]

For many years now, our hope to extend the possibility of bone marrow grafting to patients without compatible siblings has been based on progress in the tissue-typing field. In view of the practical obstacles in obtaining sufficiently large series of patient data from attempts at introducing selected nonsibling donors, preclinical experiments in experimental animals are in my opinion a prerequisite. Both the dog and the monkey are presently available as excellent models for such studies because histocompatibility testing in these two outbred species has come to the stage that sufficient knowledge and methodology are available to translate the results into terms of human tissue-typing. In the monkey and the dog LD and SD typing are now as well developed as in man.[11,12] In addition, fast progress is being made in the area of typing for Ia-like antigens in the monkey, since, in contrast to the situation in humans, it is much more feasible to produce specific typing sera by immunization of selected individuals. The observed association between Ia-like antigens and LD specificities in the monkey[11,13] (either due to a high linkage disequilibrium or to a more direct role of Ia antigens in MLC) greatly facilitates the selection of LD-compatible unrelated host–donor combinations. In the dog, a similar association exists between LD and SD determinants. Therefore, in the immediate future, pertinent information about the influence of LD and Ia-like antigens on graft take and GVHD may be expected to appear for both species. Preliminary information regarding the influence of SD and LD matching was presented at this conference by Storb et al,[14] and Vriesendorp et al.[15] In monkeys the procedures of stem-cell purification, bone marrow preservation, and bacteriologic decontamination of the recipient are all operational,[16] so that these procedures can be combined with donor selection among unrelated animals.

(2) There is a high incidence of GVHD among patients grafted with compatible sibling marrow, which is responsible for the death of 25%–50% of the patients in whom a take of bone marrow was obtained. For this discussion it is assumed that the high incidence of fatal interstitial pneumonias in leukemia patients is related to the occurrence of low-grade GVHD. Our approach to this problem in the past few years has been based on the following philosophy: a number of different techniques have become available that allow effective reduction of the incidence and severity of GVHD in experimental animals, e.g., pretreatment of the recipient with ALG, selective elimination of immune competent cells from the graft by way of stem-cell purification with BSA density gradients, by preincubation of the bone marrow with ALG,[17] or other in vitro manipulations, and finally by preventive treatment of the recipient with MTX and/or other agents.[5] All these measures carry the risk of reducing the takeability of the graft or of toxicity for the recipient. Their full application is in fact counterindicated for those recipients who will not develop GVHD anyhow. The systematic introduction of such GVHD reductive measures, therefore, requires a re-

Table 3. Investigations of Predictive Factors of GVHD in DLA-Identical Sibling Donor-Recipient Combinations in Dogs

Factor Studied	Conclusion	Reference
Specific property of donor bone marrow	No	—
Third-party disparity ratio MLC[23]	No	24
Blood transfusion (irradiated) posttransplantation 100 ml on days 7, 14, 21, 28	Incidence: no Severity: Yes	25
Red cell antigens	No	24, 25
Polymorphic red or white cell antigens	PGM2 incompatibility shows positive (<100%) correlation with GVHD*	25
ABO-like polymorphic digestive tract system[18]	CSA incompatibility shows negative (<100%) correlation wih GVHD†	25

*PGM2 is not polymorphic in humans; one case of severe GVHD described in PGM2-incompatible HLA-identical sibling transplant. Retrospective determination of PGM2 in human HLA-identical transplantation recommended.

†Could serve for donor selection in case more than one HLA-identical sibling donor is available.

liable prospective identification of the patients who will develop GVHD. It appears that the only available animal model for developing such a predictive test is the dog, because in this outbred species, large litters, as well as advanced histocompatibility typing methodology, are available. Table 3 lists the various factors so far investigated. Most parameters were found to bear no relationship to the development of GVHD, but two genetic systems, the PGM2 system and the canine secretory alloantigen system (CSA) as defined by Zweibaum and co-workers[18] showed certain indications of a relationship with GVHD. This justifies more systematic studies along the same lines in human bone marrow transplantation cases. The same applies for the two factors reported to correlate with GVHD in a series of aplastic patients treated with matched sibling marrow, namely, sex mismatch and refractivity to random platelet infusions.[9] It is, however, by no means certain that dependable prognostic factors can be found. If it would turn out that a large number of minor histocompatibility systems—instead of one or two—are of importance in the development of GVHD in the MHC-identical sibling, the severity of the GVHD would be expected to be determined by the number of incompatibilities of the minor histocompatibility systems as well as by the number of cells grafted. Obviously, it would be extremely difficult if not impossible to develop a predictive typing procedure under these circumstances.

Further progress in this area and the eventual answers will depend greatly on extension of our fundamental know-how of the processes determining the development of clinical symptoms of GVHD.

INTESTINAL GVHD

One of the most dangerous and disturbing manifestations of GVHD is diarrhea, which if persistent, may lead to wasting and death. It occurs both in the acute and in the delayed form of GVHD. In mice, the diarrhea, wasting, and mortality of the delayed type of GVHD can be prevented by total bacteriologic decontamination or by using germ-free mice.[19] Even when the normal gut flora is reintroduced in germ-free or decontaminated chimeras, at a time (e.g., 40 days) when the diarrhea and the mortality are still highly prevalent in conventional animals, the disease does not develop (Fig. 1). Acute GVHD (or acute secondary disease), which can be induced in lethally irradiated mice by adding spleen cells to the allogeneic bone marrow graft, is also significantly decreased, but the mortality is not prevented by decontamination or in germ-free animals. These findings suggest that in allogeneic mouse radiation chimeras the severity of the GVH reaction in the gut is determined by the presence or absence of microflora. The classical hypothesis was

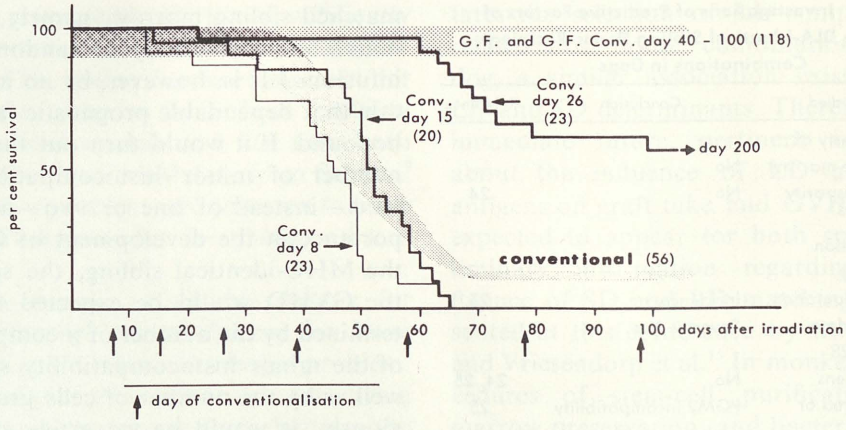

Fig. 1. Effect of conventionalization on mortality of GF CBA mice following whole-body irradiation with a dose of 900 rads x-rays and transplantation of 10^7 C57BL bone marrow cells. Shaded area 1, conventional controls; shaded area 2, GF animals and GF animals conventionalized on day 40 or later. Conv., conventionalization; GF, germfree. The methods employed have been described in detail previously.[19]

that the GVH reaction in the gut is primarily determined by histoincompatibility and subsequently aggravated by infection of the primary lesions. However, the magnitude of the beneficial effect of removal of the microflora on both the delayed and the acute GVHD suggested that other factors, such as stimulation or activation of donor type immune competent cells by components of the gut flora might play a role. To distinguish between these possibilities we used a model in which the GVHD lesions can be studied in normal and in germ-free intestines carried by the same chimeric animal. For this purpose, fetal gut implants as described by Zinzar et al.[20] were prepared in (CBA × C57BL) F1 hybrid mice, which were then irradiated and treated with CBA marrow and spleen cells so that acute secondary disease developed. At various intervals thereafter the mice were sacrificed and the lesions in the mice's own intestinal tracts as well as in the implants were scored quantitatively in histologic sections. Certain groups of mice carried both an F1 (host type) and a CBA (bone marrow donor type) fetal gut implant. Decontaminated chimeras were compared with conventional animals.[21] Surprisingly, it was found that in conventional mice the (germ-free) CBA gut implant (isogeneic with the grafted hemopoietic cells) showed a significant amount of specific damage, while in decontaminated chimeras the CBA implant was free of histologic lesions (Table 4). Clearly, the presence of microflora at a distant site is capable of magnifying the damage in the (sterile) F1 gut implant and to induce damage in the CBA implant, which is histocompatible with the cells that induce the primary GVHD lesions. Therefore, a new hypothesis is required: It postulates the presence of antigens on certain intestinal bacterial species which cross-react with antigens of intestinal epithelial cells. The initial GVH reaction is directed against histocompatibility antigens only. When the resulting lesions become infected, microorganisms may penetrate into the body and if these microorganisms carry antigens cross-reacting with gut epithelium, addi-

Table 4. Crypt Damage (%) in Fetal Gut Implants of Mice With Acute Secondary Disease

Chimera D → H	Fetal Gut Source F1	Fetal Gut Source CBA
CBA → F1 *conventional*	47	23
CBA → F1 *decontaminated*	23	3
F1 → F1 *conventional*	<3	<3
F1 → F1 *decontaminated*	<3	<3

tional lymphocytes may be activated to react against the intestinal epithelium, thus reinforcing the initial damage. Obviously, a similar mechanism could be proposed to explain the GVHD in the skin, which appears also to be less predominant in germfree animals as compared to conventional mice.

The observations on conventionalization of germ-free chimeras indicate that the period of initial intestinal damage induced by the GVHD reaction per se, is limited to the first month after allogeneic bone marrow transplantation and that the lesions heal without sequelae in the absence of microorganisms. In the presence of a microflora, however, reinforcement results in more widespread damage. On top of that, chronic ulceration of both types of lesions may occur, which seems to be the predominant pathologic process during the third month of the posttransplantation period. The proposed mechanism requires the identification of bacterial species that cross-react with antigens of intestinal epithelium, and the demonstration that the selective removal of these bacterial species from the microflora will result in significant reduction of the intestinal manifestation of secondary disease. This mechanism may also throw some light on the negative correlation obtained in dogs between the incompatibility for the CSA antigens of the digestive tract and the occurrence of GVHD. Such incompatibility may correlate with the presence of antibodies against the recipient CSA antigens. These antibodies might cross-react with certain bacterial antigens, thus preventing their penetration through the intestinal lining. The expected frequency of W/Z antigen incompatible siblings in the human is at least in the range of the GVHD incidence (Table 5). Admittedly, these relationships are based on speculations only, but they serve to encourage the clinical transplantation teams to collect more information on donor and host digestive tract antigens and on the composition of the microflora of

Table 5. Antigens of Secreting Cells of the Human Digestive Tract

Antigens	Gene Frequency
W	0.37
Z	0.59

Computed frequency of W/Z incompatible siblings is 45%.

Alloantibodies to the antigen(s) absent in an individual are present in his serum.

Methodology: immunofluorescence test on rectal biopsy material.

Adapted from Zweibaum and Oriol.[18]

transplantation patients. With the continued introduction of isolation and decontamination technology in bone marrow transplantation clinics the required bacteriologic information should be easily accessible.

SUMMARY

Improvements in the results of bone marrow transplantation for the treatment of SCID may be expected by employing purified stem-cell concentrates for patients who do not have a compatible sibling available. Refinements in the purification technique and its monitoring are required, however. For the same category of patients it seems worthwhile to continue attempts at restoration with liver cells from fetuses less than 12 weeks of age. In addition, full protection against infections should be provided for patients expected to develop GVHD, and, therefore, such patients should only be treated in centers where reverse isolation and bacteriologic decontamination can be performed. In view of the rarity of the disease, transplanters should agree on a limited number of graft protocols.

For the treatment of bone marrow aplasia, attempts to identify the factors that can serve to predict the occurrence of GVHD in compatible host-donor sibling pairs should be continued. Only when the patients who will develop GVHD can be recognized in advance will it be feasible to fully exploit available GVHD reductive

measures. In particular the role of the intestinal microflora should be investigated in this respect. Experimental evidence is presented, suggesting an aggravating influence of microflora on GVHD lesions, which are primarily induced by histocompatibility reactions. For such studies with incompatible siblings, the dog is the best available animal model. For the selective isolation of hemopoietic stem cells for transplantation purposes (as one means of reducing GVHD), methods for rapid identification of stem cells and immune competent cells, respectively, have to be developed.

In leukemia, more research is necessary on the factors that play a role in the late complications of bone marrow transplantation. The toxicity of aggressive regimens employed in the eradication of the leukemia should be further analyzed. The collection of autologous normal hemopoietic stem cells from leukemic patients as introduced by Dicke et al.[22] warrants further exploration to see whether these cells may replace the allogeneic transplantation procedure, thus avoiding all the complications generally encountered in GVHD.

For all three diseases, it is extremely important to develop a method for the selection of compatible donors among unrelated individuals, because this will at least double the number of candidates for therapeutic bone marrow transplantation. Current progress in histocompatibility typing in the rhesus monkey and the dog makes these species excellent models for such investigations.

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